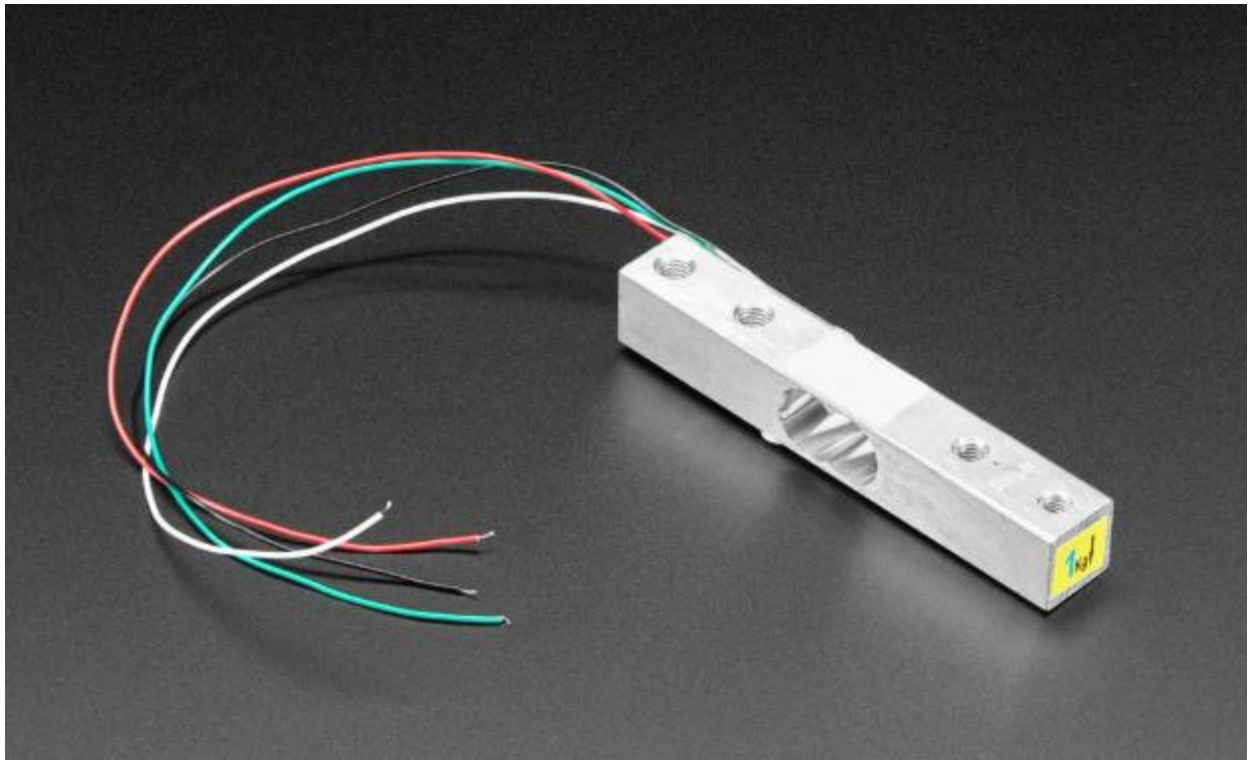


Load Cell Transducer



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Under the Instructions of:

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Zewail City of Science and Technology

Fall 2025

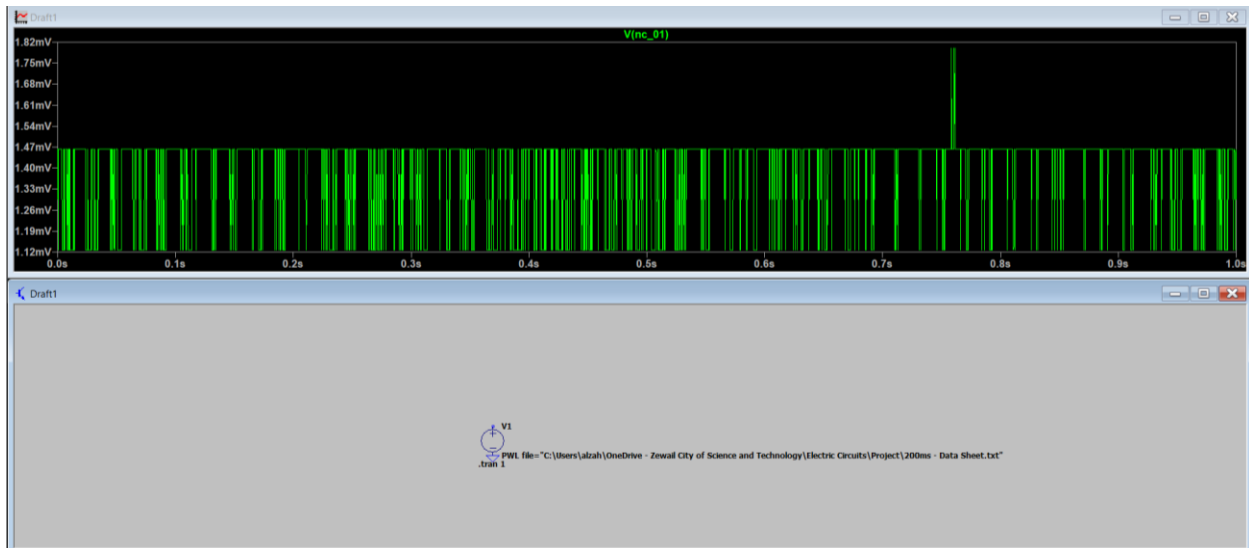
Stage 2

What we did:

- We started this stage off by creating our circuit after getting the required components and connected the circuit to Analog Discovery.
- We were able to get our data and analyze the signals on Waveforms, plotting the collected data back in LTspice to analyze the effect of load on our load cell in certain bases.
- The data Collected at base of 2ms:
https://drive.google.com/file/d/1lBZykyYBvNmbow2Tnp_mRkUnbqFRG_V-U/view?usp=sharing
- The video of Signal behavior at base of 2ms:
https://drive.google.com/file/d/1cMXPdRDqaMkzonC2-n0lnNpfiiKaPJhz/view?usp=drive_link
- The data Collected at base of 200ms:
https://drive.google.com/file/d/1bSerucw6JmdWpj3PwJVzd_wtx0cJ8PII/view?usp=drive_link
- The video of signal behavior at base of 200ms:
https://drive.google.com/file/d/1iQzKSM5lXHc_dUd_0DKo62kIPhYNEtrQ/view?usp=drive_link

- Analyzing the videos, we can see that the behavior of the signal changes when load is applied to the sensor, indicating that the connection of our circuit is correct.
- We noticed that the direction of pressure applied to the sensor is important, as the force applied can turn into components in each direction (Horizontal and Vertical), so we made sure to apply the correct pressure on the sensor to get the best results.
- Another thing is how we connected the sensor to the breadboard; it was important to connect it correctly as noise was noticed excessively when the wires were mostly covered, so we made sure to fix that.
- The sensor exhibits continuous high-frequency fluctuations and a small drift in its DC baseline; the system requires proper low-pass filtering.
- From the videos we can see that when the pressure is applied, the signal rises up until it becomes a horizontal line when the pressure doesn't vary, then rises up again as more pressure is applied.

LTspice Data Analysis:



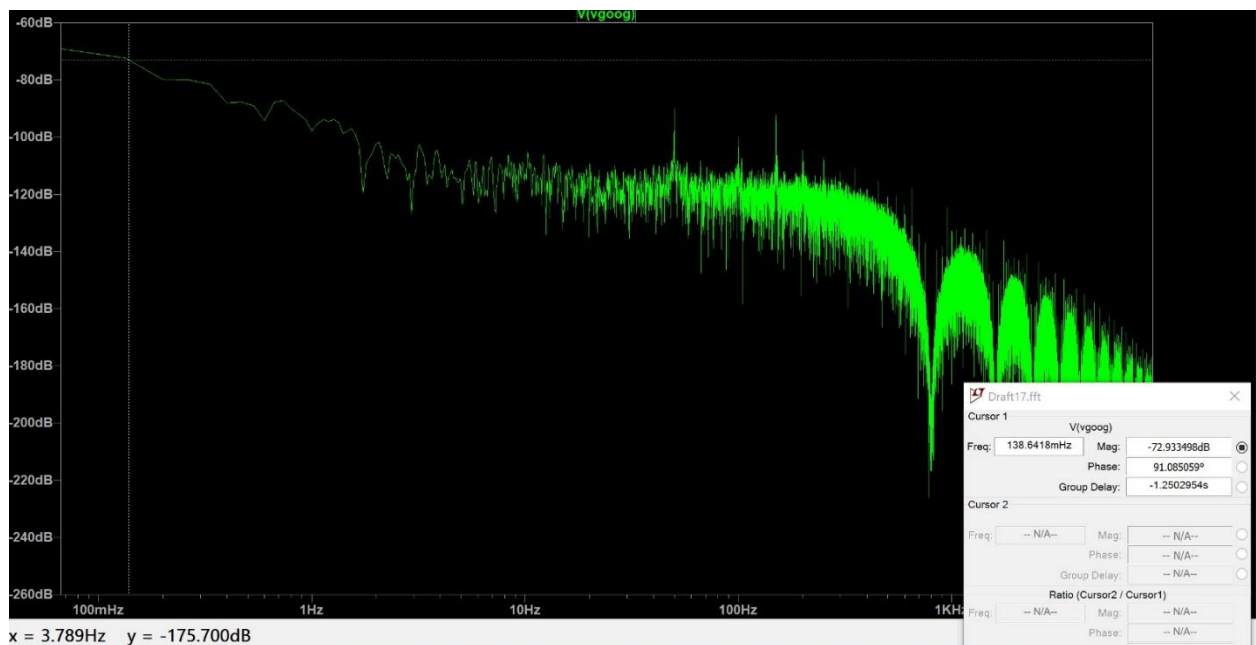
- Above is the Data Analysis that we made on LTspice according to the attached video:
<https://drive.google.com/file/d/1-Kve7wqCWil8to57lh-fNOu5F5nBVxkN/view>

Note: The result is raw, no filters or averaging applied.

- We notice that the output is dominated by noise which needs low-pass filtering.
- Since load cells are sensitive to thermal noise, mechanical micro-vibration, electrical interference, and power supply ripple, the results we got are normal.
- A low-pass filter is essential to remove the high frequency jitter.
- The overall mean value shows long-term stability indicating that the sensor itself is functioning.

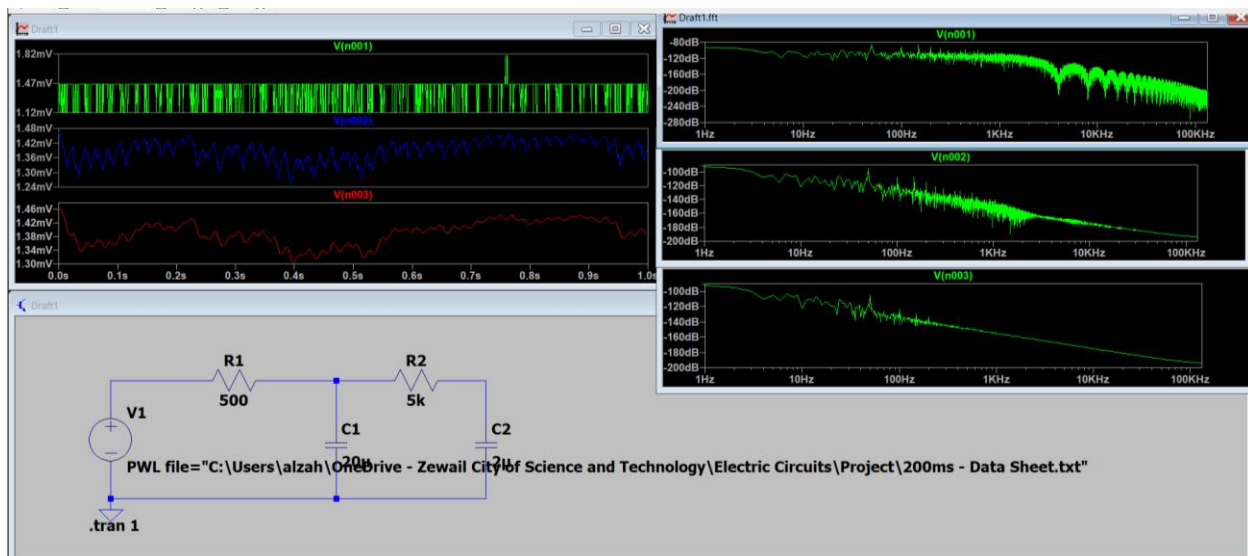
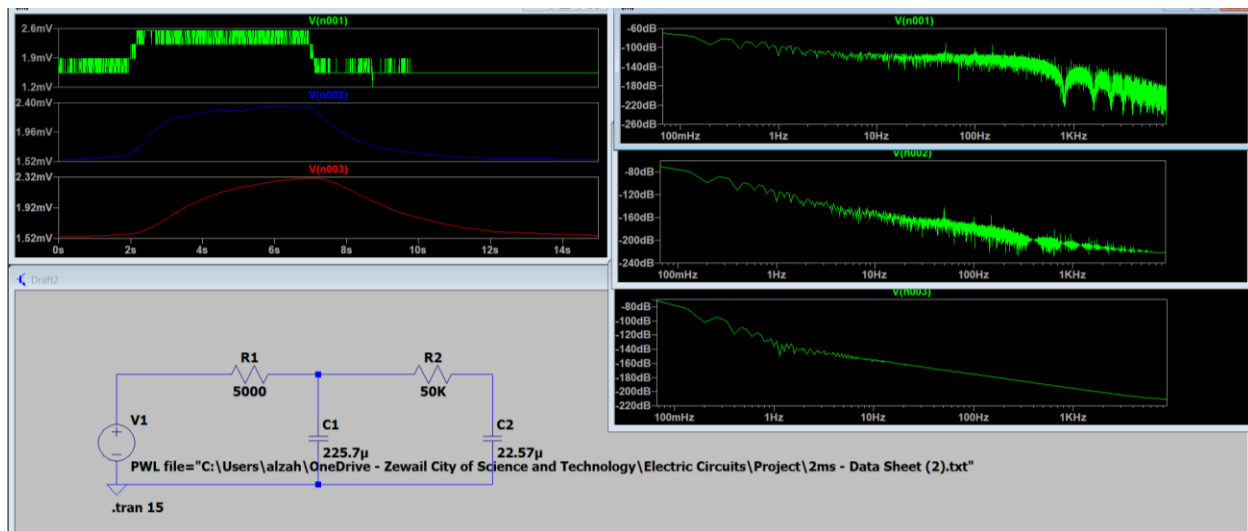
Summary of Data Analysis:

- The raw voltage waveforms obtained from the load cell show a small signal output, which is expected from a strain-gauge sensor before amplification. The signal shows high frequency noise, plus occasional spikes, which are consistent with environmental noise and mechanical micro-disturbances. While the mean voltage is stable over long time, these short-term fluctuations are indicative of the need for proper signal conditioning. Amplification and low-pass filtering will be required to achieve a stable and useable load measurement from the sensor.



This shows the Frequency cutoff

LTspice Circuit Analysis:



- Above is our analysis of the data inserted into the circuit with low-pass filters to filter noise, smooth the output, remove high-frequency interference, and produce a slowly varying signal proportional to force.
- The circuit behaves exactly like a two-stage RC low-pass: $R1 = 500\Omega$ with $C1 = 20\mu F$, and $R2 = 5k\Omega$ with $C2 =$

2 μ F. And R1 = 5K Ω with C1= 225.7 μ F, and R2 = 50K Ω with C2 = 22.57 μ F.

- The FFT shows (in n002 and n003) nearly complete removal of high-frequency content.
- The time-domain results confirm proper filtering behavior.
 - Node n001 shows large noise from the PWL excitation.
 - Node n002 shows moderate smoothing.
 - Node n003 (the output) is heavily smoothed with very slow response due to the low cutoff frequency.
- The current LTspice results confirm that the circuit is a cascaded two-pole passive low-pass which heavily attenuates frequencies above a few Hz; the output is therefore very smooth.

Stage 2 Report:

https://drive.google.com/file/d/1q1BhmOoy89-auXsODW4EV9orjTTpiZ_3/view?usp=drive_link

Link to everything:

https://drive.google.com/drive/folders/1PjnTA8GA8b8U_wOX9b5B9bUmk9lzyzzf?usp=sharing